Jian Feng

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Education

University of Waterloo Sep 2023 - Aug 2027
Bachelor's degree, Computer Science, Finance, and Statistics *Waterloo, ON, Canada*

Skills

Markets & Research: Equity research, company/industry due diligence, financial statement analysis, valuation, DCF, IRR/MOIC
Market Coursework: Fixed income, derivatives, global markets
Tools: Advanced Excel, VBA/macros, Bloomberg Terminal, Python, R, SQL, Power BI

Experience

Manulife Financial Corporation Toronto, ON, Canada
Software Engineer *May 2026 – Aug 2026*

- Improved operational reliability for an internal financial-services platform used by thousands of developers; automated **1K+ daily cross-system operations** with a **99% success rate** using request pre-validation and failure handling, surfacing rare edge cases for follow-up.
- Secured platform integrations by replacing plaintext credentials with **GitHub App token exchange** and team-scoped permissions; integrated live AKS and certificate checks into a Foundry agent to flag GitOps deployment mismatches.

Treasury Board Secretariat Toronto, ON, Canada
Software Developer *Sep 2025 – Dec 2025*

- Improved data quality in a **5K-firm public dataset** by automating duplicate and invalid job-title cleanup, reducing manual processing time **85%** and enabling peer-group and governance analysis.
- Optimized analyst reporting queries to cut load time **30%**; migrated **30 tests across five suites** to Katalon, adopted by **two QA teams** after a demo.

GuangDong Tiptop Asset Management Co., Ltd China
Analyst *Jan 2025 – Apr 2025*

- Screened **100–300+ companies** from sourcing lists, databases, reports, and founder pitch decks; advanced **20–50** to initial business-model reviews, founder calls, and project profiles.
- Supported deep diligence on **3–8 companies** and **1–3 IC-stage opportunities**, analyzing industry/competitive landscape, cohort and retention data, and three-to-five-year DCF/IRR/MOIC scenarios; contributed to IC materials and sell/hold recommendations, including a retained-stake proposal.

Lansi Institute of Artificial Intelligence in Medicine Remote
Data Analyst *May 2024 – Dec 2024*

- Built a reproducible analysis pipeline for **100GB of multi-omics data**, cutting preprocessing time **89%, from days to hours**, for an international research collaboration.
- Applied XGBoost and SMR/MR² across a **68-study meta-analysis** to screen evidence and narrow candidate drug targets for published research.

Projects

Most Efficient Portfolio | Canadian Home Bias

- Built a CAD-denominated mean-variance frontier from five regional equity ETFs over **275 months**; quadratic programs and a $\gamma = 3$ utility criterion put the optimal Canadian weight at **48%**, close to the **~50%** observed home bias versus Canada's **~3%** global market cap.
- Ran sensitivity across $\gamma = 1$–10 and the minimum-variance criterion, documenting corner solutions and error maximization.

Earnings Yield Factor | CRSP/Compustat

- Built a point-in-time trailing E/P factor on CRSP/Compustat (**1.3M stock-months**, 2001–2024); the equal-weighted Q5–Q1 portfolio earned **18.7%/yr** (Sharpe 1.03, Newey–West t = 4.13).
- FF5 spanning leaves a **14.9%/yr alpha** (t = 5.11, $R^2 = 0.50$); value-weighting halves the premium to 9.5% (t = 1.80).

Snowball Pricer | Basket Autocallable Options

- Built a market-data-to-pricing pipeline for basket autocallables: option-chain feeds → SSVI implied-volatility surface → Dupire local-vol Monte Carlo pricing and Greeks.
- Defined a model error budget for sampling, stale market data, and model-vs-dealer pricing differences; default yfinance feed is delayed, with Questrade L1 as an optional real-time feed.

Self-Directed Stocks & ETFs | moomoo, Wealthsimple

- During 1 year of active trading in 2025, broker-displayed account returns were **+30% paper** (moomoo) and **+25% live** (Wealthsimple), calculated as (ending value − initial value) / initial value.
- Returns are account-level and not benchmark-adjusted; positions informed by Morningstar research and major news.

Achievements

- 3rd of **45 teams** (School of Accounting and Finance students), SAF Investment Research Challenge; finalist, 7th Annual SAF Strategy Case Competition; WorldQuant Brain Gold Level (**20 quality alphas; Research Consultant candidate**).
- As UX Research Hub Director, owned community growth and programming end-to-end: ran member research, secured a Google UXR guest speaker (**50 attendees**, **50 new members**), grew membership **32× (10→318)**, and managed event funding with the executive team.